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Paper Code : 21313

F-413

B.C.A. (Third Semester) Examination, 2024-25

(New Course)

Paper - BCA-303-N

Data Structure Using C

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks i.e. 14 marks symbols used have their usual meanings.

1. (a) Explain the basic terminology of data structures. What are the different types of data organizations used in programming?

   (b) Discuss the concept of algorithm complexity and time-space trade-off with examples.

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2. (a) Define arrays. Explain the representation and analysis of single and multidimensional arrays. How is the address of an element calculated in a one -dimensional array?

   (b) Write a C program to perform basic operations (insertion, deletion) on a one-dimensional array.

3. (a) Construct an expression tree for the expression (a+b*c)+((d*e+f)*g). Give the outputs when you apply inorder, preorder and postorder traversals.

   (b) Create a binary search tree for the following numbers start from an empty binary search tree. 45, 26, 10, 60, 70, 30, 40 Delete keys 10, 60 and 45 one after the other and show the trees at each stage.

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4. (a) How does quick sort algorithm works? What is the output of quick sort after the 3rd iteration given the following sequence?
24 56 47 35 10 90 82 31

(b) What are the two traversal strategies used in traversing a graph?

5. (a) Write an algorithm for Push and Pop operations on Stack using Linked list.

(b) What is a DeQueue? Explain its operation with example? https://www.mjpruonline.com

6. (a) Define a graph. Explain the different types of graphs and their representations.

(b) Describe the adjacency matrix representation of a graph. How is it used for graph traversal?

7. (a) Explain the concept of hashing. Describe different collision resolution strategies used in hash tables.

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(b) Show the result of inserting the keys 2,3,5,7,11,13,15,6,4 into an initially empty extendible hashing data structure with M=3.

8. Write short notes on any **four** of the following:

(a) Biconnected Graph

(b) Sparse Matrix

(c) Vector

(d) Two-way Header List

(e) Recursion

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Paper Code : 21313

F-413

B.C.A. (Third Semester) Examination, 2023-24

(New Course)

Paper - BCA-303-N

DATA STRUCTURES USING 'C'

Time : Three Hours] ***[Maximum Marks : 70***

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. (a) Explain the various types of data structures.

   (b) Define the term recursion. Write a program in C to display factorial of a entered number using recursion.

2. (a) What are the major data structures used in the following areas: RDBMS, Network data model & Hierarchical data model?

   (b) What are the notations used in evaluation of Arithmetic Expressions using prefix and postfix forms?

3. (a) Describe working of selection sort method. Also sort given input list in ascending order using selection sort input list- 55, 25, 5. 15, 35.

   (b) Write an algorithm to count number of nodes in singly linked list.

4. (a) Construct a binary search tree for following elements: 30, 100, 90, 15, 2, 25, 36, 72, 78, 10 show each step of construction of BST.

   (b) Write algorithm for performing push and pop operations on stack.

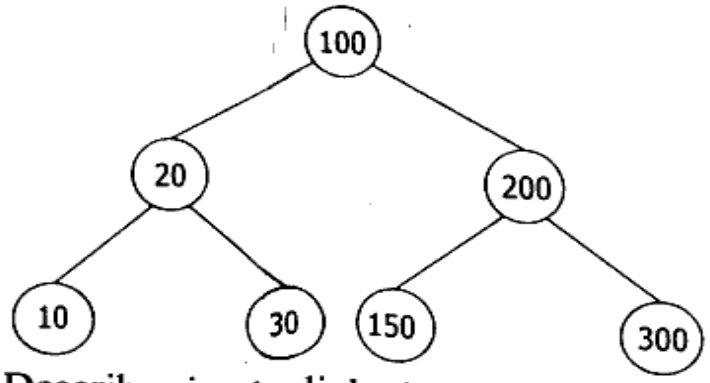
5. (a) Convert the following infix expression to its prefix form using stack A+B–C*D/E+F. Show diagrammatically each step of conversion.

   (b) Differentiate between Binary Search and Sequential Search.

6. (a) Find the position of element 29 using binary search method in an array 'A' given below : A= {11, 5, 21, 3, 29, 17, 2, 43}.

(b) Write an algorithm to delete a node from the beginning of a circular linked list.

7. (a) Traverse the following tree by the in order. Pre-order methods:



(b) Describe circular linked list with suitable diagram. Also state advantage of circular linked list over linear linked list.

8. (a) Describe working of bubble sort with example.

(b) Differentiate between general tree and binary tree.

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Paper Code : 21313

F-413

B.C.A. (Third Semester)

Examination, 2021-22

(New Course)

Paper - BCA-303-N

Data Structures Using C

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. (a) Define algorithm. How do you measure the complexity of algorithm? List the commonly used asymptotic notations. 7

   (b) Draw a binary tree using following traversals: 7

   Inorder : DBFEAGCLJHK

   Postorder : DFEBGLJKHCA

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2. (a) Write the following infix expression in corresponding postfix expression: 7

   (i) A*B/C*D

   (ii) A/B* C+d*e-A*C

   (b) What is recursion? Write a recursive 'C language' function to print fibonacci series up to 'n' terms. 7

3. (a) Write an algorithm to implement insertion sort. Illustrate this for following list of numbers: 7

   100,15,85,25,5,40,45,35

   (b) What is binary search tree? Build binary search tree using following integers.7

   57,25,65,20,35,70,80

4. (a) Write an algorithm to insert a node in the beginning, in the end, and in between the nodes of a linked list. 7

   (b) What is hashing? What are its significance and advantages? 7

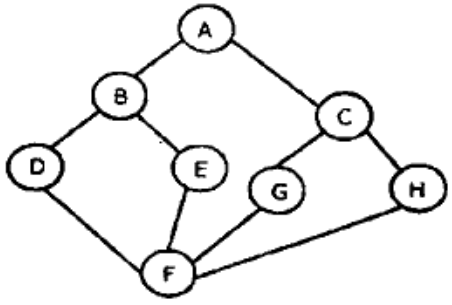
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5. (a) Write the algorithm for binary search. When is linear search preferred over binary search. 7

(b) Define circular queue. What is the condition when the circular queue is full when implemented using array? 7

6. (a) Write the algorithm for multiplication of two matrices. 7

(b) Differentiate between DFS and BFS. If 'A' is the starting vertex, find out the DFS & BFS traversal of following graph: 7



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7. (a) Write a program to implement Bubble sort. 7

(b) Sort the following sequence using bubble sort showing all steps: 7

77,33,44,11,88,22,66,55

8. Write a short note on any **four**: 14

(a) Time space trade off.

(b) Garbage collection.

(c) Comparison of indexing & hashing.

(d) Push & pop in stack.

(e) Polynomial representation using linked list.

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F-413

B.C.A. (Third Semester) Examination, 2021

(New Course)

Paper - BCA-303-N

DATA STRUCTURES USING C

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. (a) What is data structure, differentiate between non-linear data structure and linear data structure with appropriate example?

   (b) What is an algorithm? Write various asymptotive notation for algorithm complexity?

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2. (a) Define array. What are its various storage techniques in memory?

   (b) Define sparse matrix?

   (c) How can on pass array as parameter in C program, explain?

3. (a) What are queues, how can one perform addition and deletion in queue, explain with appropriate algorithm?

   (b) What are dequeues, explain?

4. (a) Write a program to create singly linked list and display the list?

   (b) Write a program to impement quick sort and show step by step for following element by quick sort :

   19 27 5 9 86 45

5. (a) What are different methods for tranersing the graph? Explain DFS in detail with an examples?

   (b) Distinguish between linear and binary search methods?

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6. (a) Write the result of evaluating the post fix expression

AB - CD*/ given

A = 2,

B=10,

C=4

D=1

(b) What are hash functions, explain with suitable example?

7. What are Binary Search Tree (BST) how can one perform insertion and deletion in binary search tree explain?

8. Write short notes (Any two) :

(i) Adacency matrix

(ii) Bubble sort

(iii) Sparse matrix

(iv) Multigraph

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F-413

B.C.A. (IIIrd Semester) Examination, 2019-20

(New Course)

DATA STRUCTURES USING C

Paper-BCA-303-N

Time : 3 Hours] [Maximum Marks : 70

Note :– Attempt any *five* questions. All questions carry equal marks.

1. (a) What is data structure ? Discuss the different types of data structures with features.

   (b) What is Stack ? Write an algorithm to implement stack using linked list.

2. (a) Explain why circular queue is better than linear queue.

   (b) Write a function to delete a node from beginning of double linked list.

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3. (a) Write a recursive function to count number of nodes in tree.

   (b) Convert the following infix expression into post-fix expression.

   (A + B – C * D)/H

4. (a) Compare Selection sort and Merge sort.

   (b) Define Hashing. Discuss any *two* hashing methods with example.

5. (a) Explain height balance tree and AVL. tree with suitable example.

   (b) Explain the concept of asymptotic notations. Explain the different asymptotic natations.

6. (a) Construct a binary tree for the following inorder traversal :

   B G H D A E C I J K F J

   (b) Explain Overflow and Underflow condition on stack.

7. (a) What is BST ? Explain algorithm for insertion in BST.

   (b) What is Graphs ? Explain Multigraphs and Directed Graphs

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8. Explain any *four* of the following :

(i) D-queue and Priority queue

(ii) Doubly Linked list

(iii) Tower of Hanoi problem with *three* disc

(iv) Sequential and Binary Search

(v) Sparse Matrix.

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B.C.A. (Third Semester) Examination, 2018

(New Course)

Paper-BCA-303-N

DATA STRUCTURES USING C

Time : 3 Hours] [Maximum Marks : 70

Note :– Attempt any *five* questions. All questions carry equal marks.

1. (a) Define Data Structures. What is the difference between linear and non-linear data structures ? What are the operations that can be performed on data structures ?

   (b) Discuss the array representation of stacks. Write the algorithm for push and pop stack operations.

2. (a) What is a Sparse Matrix ? Describe a data structure for the efficient storage of a sparse matrix.

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   (b) Write the postfix form of each of the following infix expressions :

   (i) A – B + (M $ N) * (O + P) – Q/R^S * T + Z

   (ii) K + L – M * N + (O ^ P) * W/U/V * T + Q

3. (a) What is Queue ? How is queue different from a stack ? Discuss the various operations on queues.

   (b) What is a priority queue ? How can it be implemented ? Explain an application of priority queue.

4. (a) What is doubly linked list ? What are its applications ? Explain how an element can be deleted from the list using appropriate pseudo code. http://www.mjpruonline.com

   (b) Write a program to create a linked list of names in such a way that after every insertion the list is always in sorted order.

5. (a) Define a binary tree. What do you mean by tree traversal ? Using the following traversal construct the corresponding binary tree :

   INORDER : H K D B I L E A F C M J G

   PREORDER : A B D H K E I L C F G J M

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(b) What is Hashing ? Explain various methods for resolving hash collisions.

6. (a) What is run time complexity of an algorithm ? Calculate the run time complexity of bubble sort.

(b) Explain Quick Sort with the help of suitable example.

7. (a) What is Binary Search Tree ? Write an algorithm to delete an element D from a binary search tree so that the tree remains a binary search tree.

(b) What do you mean by graph traversal ? Define depth-first traversal (DFS) of a graph. Write an algorithm of non-recursive depth-first traversal.

8. Write short notes on any *four* of the following :

(i) Complete Binary Tree

(ii) Hash functions

(iii) Adjacency matrix representation of graph

(iv) Circular Queues

(v) Overflow and underflow in linked list.

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